Shujun Zhao

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Overview

Shujun Zhao is currently a Ph.D. candidate under Dr. Xiang-dong Hou's supervision in the Department of Mathematics and Statistics at the University of South Florida. She has passed the qualifying exams in Real Analysis (May 13, 2023) and Algebra (August 15, 2023) at Ph.D. level, as well as the specialty qualifying exam on Finite Fields (November 2, 2024). Her current research interests include function fields over finite fields, Drinfeld modules, and polynomials and rational functions over finite fields with applications in coding and cryptography.

Education

University of South Florida — Tampa, FL
Ph.D. Candidate, Mathematics. — *Aug 2022 – Aug 2027(Expected)*

Guilin University of Electronic Technology — Guilin, Guangxi
B.S., Information and Computing Science. — *Sep 2018 – Jun 2022*

Publications

1. Xiang-dong Hou, **Shujun Zhao**. "On a Conjecture About the Sum-Freedom of the Binary Multiplicative Inverse Function." *Designs, Codes and Cryptography 94.4 (2026): 86.*
2. Alyssa Ebeling, Xiang-dong Hou, Ashley Rydell, **Shujun Zhao**, "On Sum-Free Functions", *Finite Fields and Their Applications 110 (2026): 102744.*
3. Xiang-dong Hou, **Shujun Zhao**, "Two Absolutely Irreducible Polynomials over $\mathbb{F}_2$ and Their Applications to a Conjecture by Carlet", *Finite Fields and Their Applications 109 (2026): 102713.*
4. Luca Bastioni, Giacomo Micheli, **Shujun Zhao**. "On the Characteristic Polynomial of Linearized Polynomials." *accepted by Mathematics of Computation.*
5. Mohamed O. Darwish, Anuvrat Jaindungarwal, Giacomo Micheli, Divyesh Vaghasiya, **Shujun Zhao**, "Good Polynomials for Locally Recoverable Codes: Classification Results, Asymptotics, and Exact Formulae", *Journal of Algebra and Its Applications (2025): 2541017.*
6. Xiang-dong Hou, **Shujun Zhao**, "Further Results on Sum-Freedom of Binary and q-ary Functions" *arXiv:2609.31489.*
7. Xiang-dong Hou, Siyu Peng, Yongyu Qiang, **Shujun Zhao**, "Classification of Rational Functions of Degree Three over Finite Fields" *arXiv:2604.18954.*

Research Experience

University of South Florida — Tampa, FL

- **REU Site: Cryptography and Coding Theory** — *May 2025 – Aug 2025*
  Faculty supervisor: Xiang-dong Hou
  Contributions: Research Assistant and graduate mentor, focusing on classification of all rational functions of degree 3 over finite fields of odd characteristic.
- **REU Site: Cryptography and Coding Theory** — *May 2024 – Aug 2024*
  Faculty supervisor: Xiang-dong Hou
  Contributions: Research Assistant and graduate mentor, focusing on Carlet's conjecture concerning the sum-freedom of the binary multiplicative inverse function.

Activities

Invited speaker
On sum-freedom of binary power functions, 2027 Joint Mathematics Meetings (JMM 2027) AMS Special Session on Developments in the Theory of Boolean Functions, Chicago *January, 2027*
On a conjecture about the sum-freedom of the binary multiplicative inverse function, AMS 2026 Spring Eastern Sectional Meeting, Special Session on Finite Fields and Their Applications, Boston *March, 2026*

Conferences
Classification of rational functions of degree three over finite fields(Presented online), The 11th International Workshop on Boolean Functions and their Applications (BFA), Sogndal *September, 2026*
On a conjecture about the sum-freedom of the binary multiplicative inverse function(Presented online), The 10th International Workshop on Boolean Functions and their Applications (BFA), Cyprus *September, 2025*

Summer Schools and Professional Development
Graphical Models in Algebraic Combinatorics, Simons Laufer Mathematical Sciences Institute(SLMath) Nominated by the Department of Mathematics and Statistics, University of South Florida *Jun 2025 - Jul 2025*

Peer Review
Reviewer for Finite fields and their applications. *journal ISSN: 1071-5797*
Reviewer for Journal of Number Theory. *journal ISSN: 0022-314X*

Award

M.V. Johns Jr. Scholarship for Graduate Study in Statistics University of South Florida
College of Arts and Sciences *2025 - 2026*

Fred L. and Helen Tharp Endowed Scholarship University of South Florida
College of Arts and Sciences *2024 - 2026*

Teaching Experience

University of South Florida Tampa, FL
Primary Instructor *Jan 2026 - Dec 2026(Expected)*

Spring&Fall 2026	Calculus II (Integration techniques and their applications, sequences and series, parametric and polar curves)

Teaching Assistant *Aug 2022 - Aug 2026*

Summer 2026	Introduction to Probability
Fall 2025	Vector Calculus | Advanced Linear Algebra I (Graduate Level)
Spring 2025	Algebra II (Graduate Level) | Vector Calculus | Engineering Calculus III
Fall 2024	Algebra I (Graduate Level) | Engineering Calculus III
Summer 2024	Precalculus Algebra/Trigonometry
Spring 2024	Elementary Abstract Algebra I | Vector Calculus
Fall 2023	Engineering Calculus III | Elementary Abstract Algebra I
Summer 2023	Precalculus Algebra/Trigonometry
Spring 2023	Engineering Calculus II | Intro to Discrete Structures
Fall 2022	Calculus I | Calculus III

Skills

Computer Skills: Proficient in Wolfram Mathematica, MATLAB, LaTeX, and familiar with C/C++.
Languages: Native in Mandarin, proficient in English.